

Positive-Entropy Systems Admit Divergent Non-Commuting Double Averages

Candidate partial solution packet
Run fa_banach_001 — lane 0

9 August 2026

Status: candidate partial result, likely valid. Austin’s question has an affirmative answer for every weakly mixing probability-preserving automorphism of positive entropy. More generally, the proof needs only ergodicity and positive entropy. A subsequent identification shows that the universal affirmative question is false: Chacon’s weakly mixing transformation forces convergence against every second system. The theorem here remains a positive-entropy occurrence result.

1 Source problem

Let S and T be invertible probability-preserving transformations of a standard probability space $(X, \mathcal{B}, \mu)$, and let $f \in L^\infty(\mu)$. Austin [1] studies the scalar averages

$$A_N(S, T; f) := \frac{1}{N} \sum_{i=0}^{N-1} \int_X f(S^i x) f(T^i x) d\mu(x). \quad (1)$$

His examples include isomorphic rigid Gaussian transformations S, T for which these averages diverge. The conclusion of the paper asks the following broader question (arXiv PDF page 6).

6

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The known examples of non-convergence of the averages (1) provide rather diverse measure-preserving transformations in the role of S (or T). Many further questions about such transformations could be pursued. For example, it would be interesting to know whether *any* weakly mixing transformation can appear as S ; whether the ranks of S and T could help to control the averages in (1) (Gaussian examples always have infinite rank, and so do those in [14]); or which two-step distal transformations can appear.

In [8], Frantzikinakis and Host also pose various questions about recurrence involving non-commuting transformations. I do not think the methods of the present note come close to answering these.

Acknowledgements. I am grateful to Mariusz Lemańczyk for bringing recent work in this

Source question. *Can every weakly mixing transformation appear, up to isomorphism, as S in a non-convergent example of the averages (1)?*

The theorem below settles every positive-entropy instance. It says nothing about zero entropy; a separate literature-implied obstruction now shows that no uniform affirmative extension to all weakly mixing systems is possible.

2 Proof intuition

The weak Pinsker theorem [2] decomposes every positive-entropy ergodic system into a direct product of a nontrivial Bernoulli shift and a remainder. The only new construction needed is therefore on an arbitrary Bernoulli shift.

On a two-sided Bernoulli shift, a permutation p of the coordinate labels induces a measure-preserving coordinate permutation C_p . The conjugate $\tau = C_p^{-1}\sigma C_p$ is another copy of the same Bernoulli shift. We choose p so that its fixed points among $0, 1, \dots, N-1$ have no asymptotic density. If f is a centered observable of coordinate zero, then at time i the two copies of f inspect coordinates i and $p(i)$. Their correlation is the variance of f when those coordinates coincide and zero when they are distinct. Thus the correlation averages are exactly a constant multiple of the non-convergent fixed-point densities. No approximation is involved.

3 The theorem

Theorem 1. *Let $(X, \mathcal{B}, \mu, S)$ be an ergodic invertible probability-preserving system with positive Kolmogorov–Sinai entropy $h_\mu(S) > 0$, allowing $h_\mu(S) = \infty$. Then there are an invertible probability-preserving transformation T of X and a bounded real function $f \in L^\infty(\mu)$ such that:*

1. *T is measurably isomorphic to S ; in fact, T is conjugate to S on the same probability space;*
2. *the sequence $A_N(S, T; f)$ in (1) does not converge.*

More precisely, f may be chosen so that

$$\liminf_{N \rightarrow \infty} A_N(S, T; f) = 0, \quad \limsup_{N \rightarrow \infty} A_N(S, T; f) = \|f\|_2^2 > 0.$$

Consequently, Austin’s question has an affirmative answer for every weakly mixing transformation of positive entropy.

We first isolate the elementary Bernoulli construction.

Lemma 2. *There is a permutation $p : \mathbb{Z} \rightarrow \mathbb{Z}$ with $p(0) = 0$ for which*

$$d_N := \frac{1}{N} \#\{0 \leq i < N : p(i) = i\}$$

satisfies $\liminf_N d_N = 0$ and $\limsup_N d_N = 1$.

Proof. Set $N_0 = 1$. Recursively choose an even integer $L_k > kN_k$ and put $N_{k+1} = N_k + L_k$. Thus every block $I_k = [N_k, N_{k+1}) \cap \mathbb{Z}$ has even cardinality. On blocks with even k , let p be the identity. On blocks with odd k , interchange adjacent pairs, so $p(N_k + 2j) = N_k + 2j + 1$ and $p(N_k + 2j + 1) = N_k + 2j$. Finally, let $p(n) = n$ for $n \leq 0$.

These disjoint blockwise prescriptions define a permutation of $\mathbb{Z}$ fixing zero. At the endpoint of an identity block, the newest block alone occupies more than $k/(k+1)$ of all preceding nonnegative indices, so the fixed-point density tends to one along those endpoints. At the endpoint of a swapped block, all fixed points lie before that block, and hence their density is less than $1/(k+1)$. The two asserted subsequential limits follow. $\square$

Lemma 3. Let (K, κ) be any nontrivial standard probability space, let $Y = K^{\mathbb{Z}}$ with product measure $\nu = \kappa^{\mathbb{Z}}$, and let

$$(\sigma y)_n = y_{n+1} \quad (n \in \mathbb{Z})$$

be the two-sided Bernoulli shift. There are a transformation τ conjugate to σ and a bounded real $g \in L^\infty(\nu)$ for which

$$\liminf_N A_N(\sigma, \tau; g) = 0, \quad \limsup_N A_N(\sigma, \tau; g) = \|g\|_2^2 > 0.$$

Proof. Choose a nonzero bounded real function h on K with $\int_K h d\kappa = 0$; such a function exists because the base is nontrivial. Let p be the permutation from Lemma 2, and define the coordinate permutation

$$(C_p y)_n := y_{p(n)}.$$

Product measure is invariant under coordinate permutations, so C_p is an invertible measure-preserving map. Put

$$\tau := C_p^{-1} \sigma C_p, \quad g(y) := h(y_0).$$

Then τ is conjugate to σ . Since $p(0) = 0$, for every $i \geq 0$,

$$(\tau^i y)_0 = (C_p^{-1} \sigma^i C_p y)_0 = y_{p(i)}.$$

It follows that

$$g(\sigma^i y) = h(y_i), \quad g(\tau^i y) = h(y_{p(i)}).$$

If $p(i) = i$, their integral product is $\|h\|_{L^2(\kappa)}^2$. If $p(i) \neq i$, the two coordinates are independent and h is centered, so the integral product is zero. Therefore

$$\int_Y g(\sigma^i y) g(\tau^i y) d\nu(y) = \|h\|_2^2 \mathbf{1}_{\{p(i)=i\}},$$

and hence

$$A_N(\sigma, \tau; g) = \|h\|_2^2 d_N.$$

Lemma 2 completes the proof. $\square$

Proof of Theorem 1. By the weak Pinsker theorem [2], for every $\varepsilon > 0$ the system (X, μ, S) is measurably isomorphic to a direct product

$$(Y \times Z, \nu \times \eta, \sigma \times R), \quad (2)$$

where σ is a Bernoulli shift and $h_\eta(R) < \varepsilon$.

If $0 < h_\mu(S) < \infty$, choose $\varepsilon < h_\mu(S)$. Entropy is invariant under isomorphism and additive under direct products, so

$$h_\nu(\sigma) = h_\mu(S) - h_\eta(R) > 0.$$

If $h_\mu(S) = \infty$, the finite bound on $h_\eta(R)$ forces $h_\nu(\sigma) = \infty$. Thus the Bernoulli factor is nontrivial in either case.

Apply Lemma 3 to obtain τ conjugate to σ and a bounded real g with divergent scalar averages. On the product space set

$$S_0 := \sigma \times R, \quad T_0 := \tau \times R, \quad f_0(y, z) := g(y).$$

If $\tau = C_p^{-1} \sigma C_p$, then

$$T_0 = (C_p \times \text{id}_Z)^{-1} S_0 (C_p \times \text{id}_Z),$$

so T_0 and S_0 are conjugate. Since f_0 ignores the Z coordinate, Fubini's theorem gives

$$A_N(S_0, T_0; f_0) = A_N(\sigma, \tau; g)$$

for every N . The latter sequence has $\liminf$ zero and $\limsup \|g\|_2^2 = \|f_0\|_2^2$.

Finally, transport T_0 and f_0 back to X through the isomorphism in (2). Conjugacy, boundedness, and all the displayed integrals are preserved. This gives the required T and f . $\square$

4 Verification, limitations, and novelty status

The proof is exact after the weak Pinsker splitting. The two calculations a reviewer should check most closely are the coordinate identity $(\tau^i y)_0 = y_{p(i)}$ and the entropy argument ensuring that the Bernoulli factor is nontrivial. Both are expanded explicitly above. The accompanying `VERIFICATION.md` gives a separate audit.

This result is partial: weakly mixing zero-entropy automorphisms include many of the difficult rigid, rank-one, and distal-adjacent regimes motivating the source question, and the argument says nothing about them. A subsequent lane-0 packet records the decisive negative obstruction: Lesigne–Rittaud–de La Rue proved that Chacon's weakly mixing transformation is universal for weak disjointness, which forces convergence on the same-space diagonal against every second system [3, Proposition 4.2]. Thus the theorem here is a broad positive occurrence result, but it cannot extend to every weakly mixing transformation.

A bounded novelty search through 9 August 2026 checked the run registry and cheap local indexes, the exact wording of Austin's question, title/citation matches for arXiv:2407.08630, and recent related arXiv papers 2407.13741, 2410.11787, and 2411.02024. Those searches found Gaussian, Poisson-suspension, simple-spectrum, and rank-one-base constructions, but no statement of the positive-entropy reduction proved here. This is not an exhaustive literature review, so novelty confidence is moderate. Human review is recommended before treating the result as established literature.

References

- [1] T. Austin, *Non-convergence of some non-commuting double ergodic averages*, Proc. Amer. Math. Soc. **153** (2025), 1701–1707; arXiv:2407.08630.
- [2] T. Austin, *Measure concentration and the weak Pinsker property*, Publ. Math. Inst. Hautes Études Sci. **128** (2018), 1–119; arXiv:1705.00302.
- [3] E. Lesigne, B. Rittaud, and T. de La Rue, *Weak disjointness of measure preserving dynamical systems*, Ergodic Theory Dynam. Systems **23** (2003), 1173–1198; doi:10.1017/S0143385702001505.